

How Can a Statistical Agency Predict Tightness?

Pascal Michailat
<https://pascalmichailat.org/w/>

Statistical agency announces a rightness x^s

Households take x^s as given

→ Household i will buy $y_i^i(x^s) = \sigma(x^s) \left[f(x^s) k_i + \frac{\mu_i}{p} \right]$

→ Household i will visit $v_i^i(x^s) = \frac{y_i^i(x^s)}{q(x^s)}$

→ Realized rightness will be :

$$x = \frac{\sum_i v_i(x^s)}{\sum_i k_i} = \frac{\sum_i y_i(x^s)}{q(x^s) \cdot R}$$

$$x = \frac{\sigma(x^s) \left(f(x^s) k + N/p \right)}{q(x^s) k y^s(x^s)}$$

$$\frac{x}{x^s} = \frac{\sigma(x^s) \left[\underbrace{f(x^s) k}_{y^s(x^s)} + \mu/p \right]}{\underbrace{x^s q(x^s) \cdot k}_{y^s(x^s)}}$$

↑ must be 1

Statistical agency aims to make a correct forecast:

They aim to announce x^s such $x^s = x$

They announce x^S such that

$$y^S(x^S) = \sigma(x^S) [y^S(x^S) + \mu/p]$$

$$[1 - \sigma(x^S)] y^S(x^S) = \sigma(x^S) (\mu/p)$$

$$y^S(x^S) = \frac{\sigma(x^S)}{1 - \sigma(x^S)} \frac{\mu}{p}$$

$$y^d(x^S)$$

Statistical agency announces x^S such that:

$$y^S(x^S) = y^d(x^S)$$

Since x (realized) = x^S (forecast)

then

$$y^S(x) = y^d(x)$$